Practical 10: AirPassengers Time Series Forecasting

Objective

The objective is to analyze the monthly international airline passengers dataset (`AirPassengers`) and forecast the passenger numbers for the next 12 months (the year 1961).

Dataset

- **Source**: Standard R dataset `AirPassengers`.
- **Observations**: 144 (Monthly data from 1949 to 1960).
- **Units**: 1,000s of passengers.

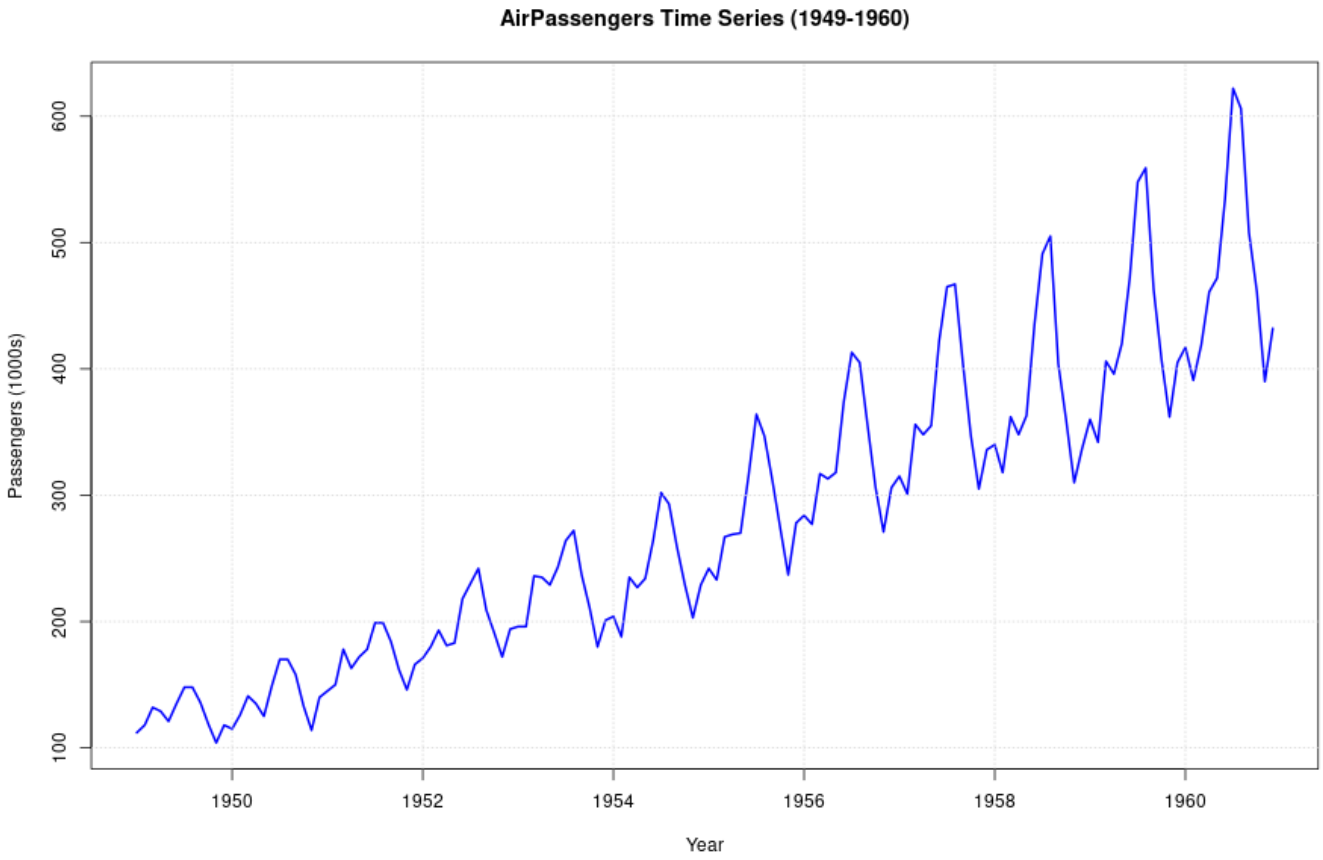
Methodology

1. **Visualization**: Plotted the original series to identify trend and seasonality.
2. **Decomposition**: Applied multiplicative decomposition because the seasonal variation increases as the trend increases.
3. **Stationarity Analysis**: Used KPSS test and ACF/PACF plots.
4. **Model Selection**: Used `auto.arima()` with a log transformation (`lambda=0`) to handle increasing variance.
5. **Validation**: Performed residual diagnostics (Ljung-Box test).
6. **Forecasting**: Projected values for the next 12 months.

Results

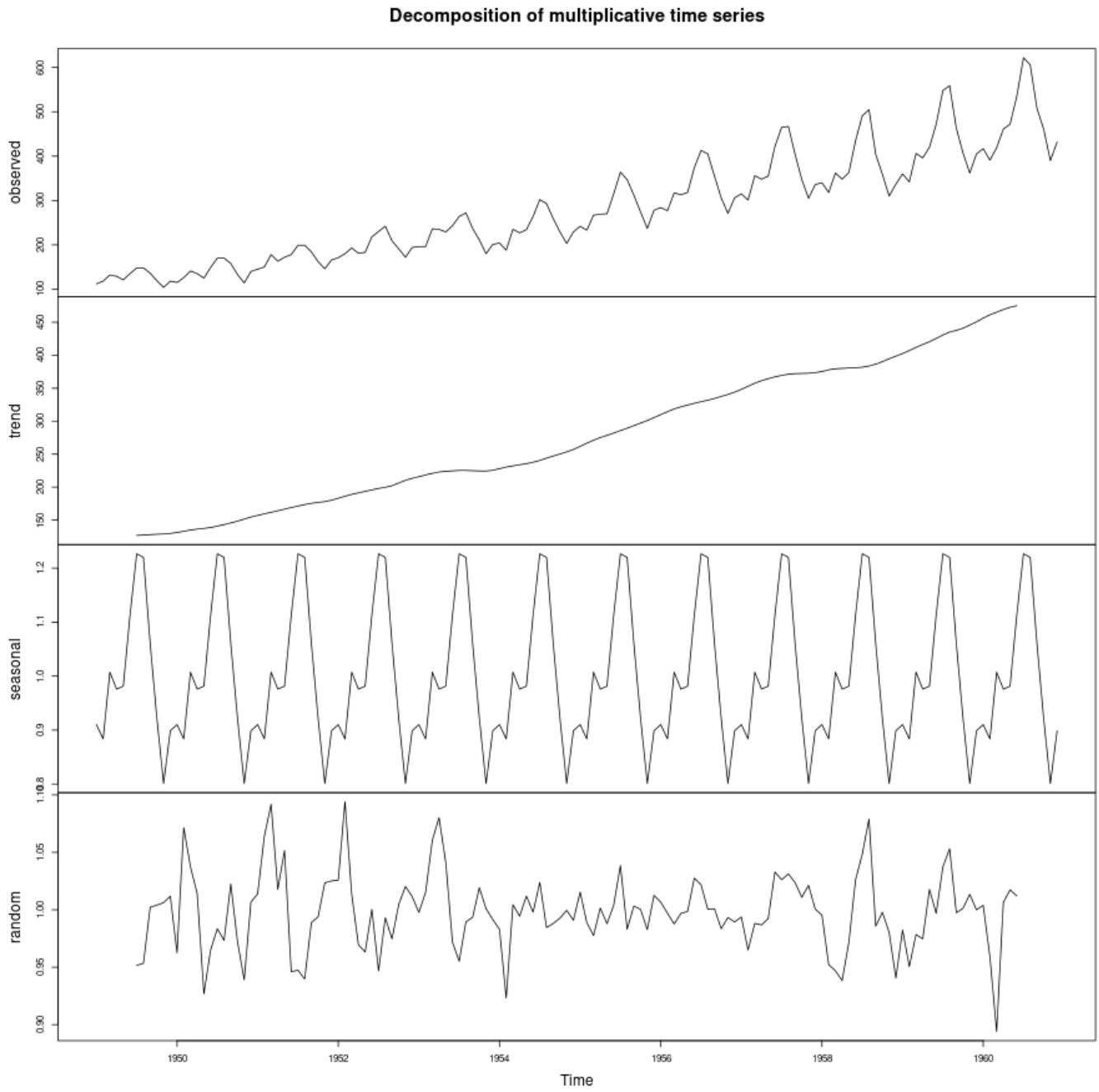
1. Data Overview

The dataset shows a strong upward trend and clear seasonality.



2. Decomposition

The decomposition confirms both an upward trend and a strong seasonal component.



3. Model Identification

The KPSS test rejected stationarity ($p < 0.01$). The `auto.arima()` function selected a seasonal ARIMA model:

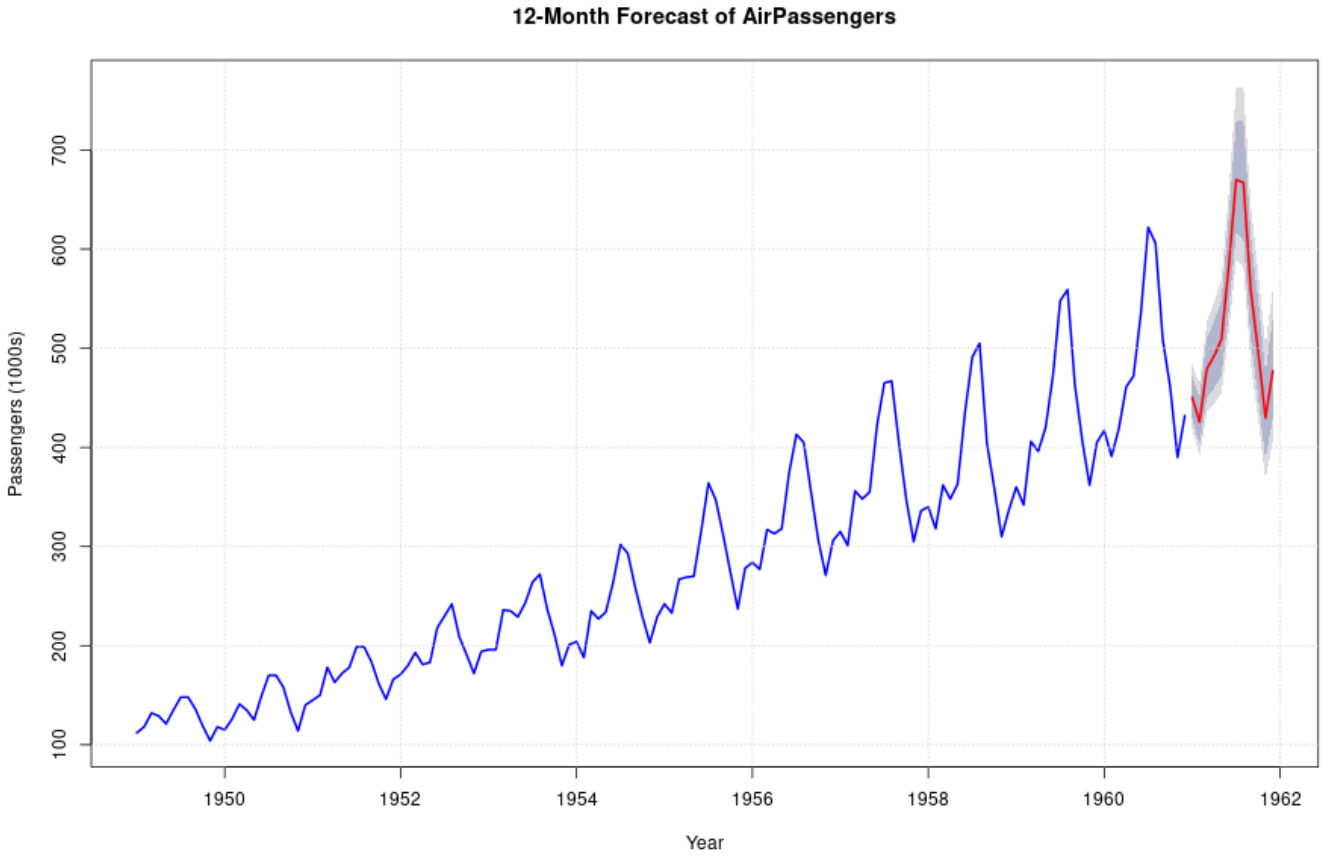
- **Model**: **SARIMA(0,1,1)(0,1,1)$_{12}$**
- **AIC**: -483.4
- **BIC**: -474.77

4. Forecast for 1961 (Next 12 Months)

The forecast continues both the growth trend and the seasonal patterns.

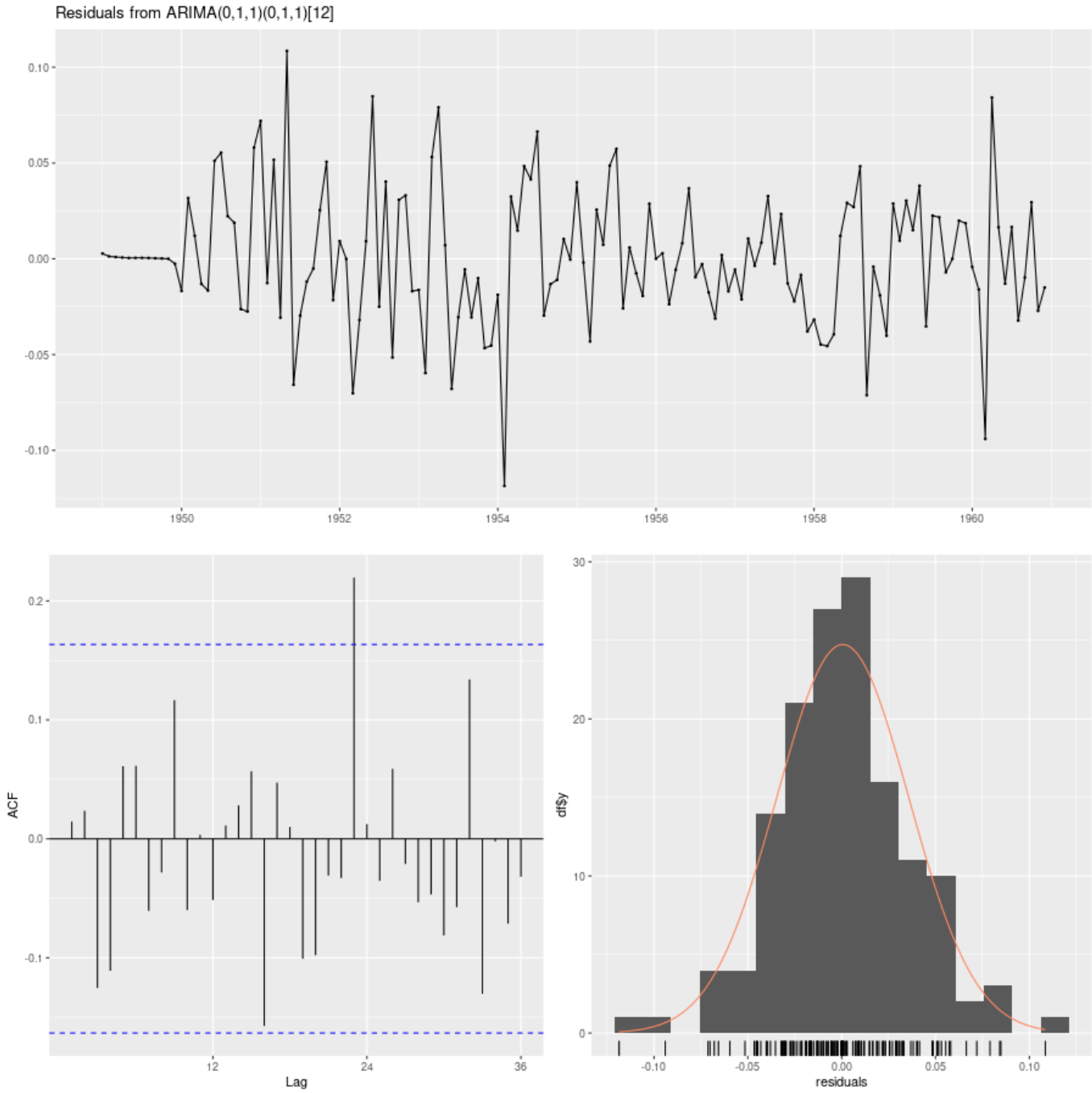
Month	Point Forecast (1000s)	95% Confidence Interval
Jan 1961	**450.4**	[418.9, 484.3]

Month	Point Forecast (1000s)	95% Confidence Interval
Feb 1961	**425.7**	[391.2, 463.3]
Mar 1961	**479.0**	[435.6, 526.8]
Apr 1961	**492.4**	[443.5, 546.7]
May 1961	**509.1**	[454.6, 570.0]
Jun 1961	**583.3**	[516.8, 658.5]
Jul 1961	**670.0**	[589.1, 762.1]
Aug 1961	**667.1**	[582.3, 764.2]
Sep 1961	**558.2**	[484.0, 643.8]
Oct 1961	**497.2**	[428.3, 577.2]
Nov 1961	**429.9**	[368.0, 502.1]
Dec 1961	**477.2**	[406.2, 560.7]



5. Residual Diagnostics

The **Ljung-Box** test gave a p-value of **0.233** (> 0.05), indicating that the residuals are randomly distributed (white noise). This confirms the model is a good fit.



Conclusion

The forecasting model **SARIMA(0,1,1)(0,1,1)$_{12}$** predicts that airline passenger numbers will continue to grow through 1961, maintaining the established seasonal peaks (peaking in July/August) and troughs (trough in November).